A Feasibility-Aware Quantum-Classical HUBO Benchmark for Occlusion-Aware UAV Trajectory Optimization

***Abstract*—Masnavi *et al.* [1] introduced the occlusion-aware target-tracking and obstacle-avoidance problem and solved it as a real-time multi-convex MPC in the continuous domain. We contribute a discrete reformulation of that problem class as a Higher-Order Unconstrained Binary Optimization (HUBO), suitable for quantum annealing and (after quadratization) QAOA. Beyond the formulation we propose a *feasibility-aware benchmarking methodology* that decomposes hard constraint satisfaction from soft trajectory quality, uses independent constraint verification, matches sampling budgets, and characterizes penalty-weight exploitation. On the $8{\times}8$ instance (960 variables, 87,684 monomials, 3,172 cubic), we benchmark classical A*, simulated annealing (SA), D-Wave neal, and Qiskit QAOA. A* and SA achieve comparable soft cost (317 vs. 303) with A* in 10 ms and SA in 47.8 s; neal is fast (0.26 s) but reaches only 22 % true feasibility, with an exploitation curve from $0\,\%$ at $\lambda{=}50$ to $60\,\%$ at $\lambda{=}2000$. QAOA reproduces the optimum on a tiny instance but does not scale. The classical baseline grounds the HUBO formulation against a conventional planner and clarifies where higher-order binary optimization becomes useful.**

***Index Terms*—HUBO, quantum annealing, QAOA, UAV trajectory, A*, feasibility, benchmarking.**

I. Introduction

Masnavi *et al.* [1] introduced the *occlusion-free target tracking with obstacle avoidance* problem for quadrotors and solved it through a multi-convex MPC; their contribution is in continuous real-time control. Recent quantum work on neighbouring problems comes from two directions: QUAV [2] addresses single-UAV obstacle-avoidance path planning with QAOA on small instances; Q4DR [3] solves drone *routing* via a hybrid QAOA-clustering + D-Wave decomposition, closer to logistics than to local motion. Neither addresses visibility-constrained trajectory planning in the quantum-optimization setting.

We bridge these: we reformulate the Masnavi problem class in a quantum-friendly discrete form and propose a benchmarking framework for annealing- and QAOA-style solvers. The contributions are: **(1)** a discrete HUBO reformulation of the occlusion-aware target-tracking class from [1]; **(2)** a feasibility-aware benchmarking methodology with a classical A* baseline; and **(3)** an analysis of why the sustained-proximity term resists QUBO flattening.

II. HUBO Reformulation

Let V be the set of grid cells and T the horizon. The decision $x[t,v] \in \{0,1\}$ is 1 iff the UAV occupies cell v at time t. The Hamiltonian has seven terms:

$$H(x) = H_{\text{uniq}} + H_{\text{start}} + H_{\text{move}} + H_{\text{obs}} + H_{\text{occ}} + H_{\text{prox}} + H_{\text{term}}. \tag{1}$$

H_{uniq}, H_{start}, H_{move} enforce kinematic constraints quadratically; H_{obs} forbids obstacle cells.

Occlusion penalty. For each cell u we trace the Bresenham line-of-sight to the target and count intervening obstacles:

$$H_{\text{occ}} = \lambda_{\text{occ}} \sum_{t} \sum_{u \in V} x[t,u]\, \big|\, \text{LOS}(u, \text{target}) \cap O\, \big|, \tag{2}$$

where O is the obstacle set. Fig. 1 shows the per-cell cost surface. For a static target this is linear; for a *moving* target it becomes a joint product of UAV and target occupancy — intrinsically higher-order.

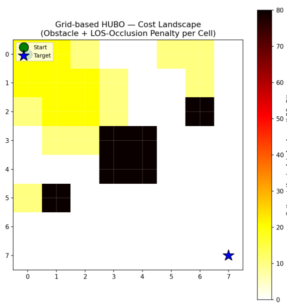

Fig. 1: Per-cell cost: obstacle penalty + line-of-sight occlusion to the target. Dark cells: obstacles. Yellow cells incur occlusion penalty because the target is hidden from them.

Genuinely-cubic proximity. The sustained-proximity term penalizes lingering in a buffer cell v_b for three consecutive steps:

$$H_{\text{prox}} = \lambda_{\text{prox}} \sum_{v_b \in B} \sum_{t=1}^{T-1} x[t{-}1, v_b]\, x[t, v_b]\, x[t{+}1, v_b]. \tag{3}$$

Eq. (3) cannot be made quadratic without Rosenberg auxiliaries — one per cubic term [4]. For our instance this adds $\sim$3,172 ancillas ($\sim$30 % more variables) with dense couplings that destroy the sparsity annealers exploit. The full instance has **960 variables**, **87,684 monomials** (3,172 cubic).

III. Feasibility-Aware Benchmarking

Raw HUBO energy is misleading because penalty-based solvers can lower energy by *violating* soft constraints. Our

TABLE I: Tier 1 head-to-head on the 8×8 occlusion-aware trajectory problem (960 variables).

Metric	A*	SA	neal
Best soft cost	317	**303**	496
Feasibility rate	**100 %**	100 %	22 %
Reaches target	**yes**	14 % runs	8 % runs
Visibility (% steps)	**80 %**	—	—
Runtime per sample	**10 ms**	47.8 s	0.26 s

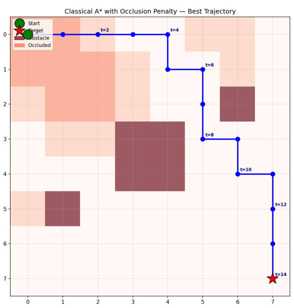

Fig. 2: A* trajectory: feasible, target-reaching, 80 % visibility.

methodology adopts four safeguards: **(i)** *energy decomposition* into hard penalty and soft cost (preventing confusion between feasibility and quality); **(ii)** *independent constraint verification* on each decoded trajectory (preventing false feasibility from energy-threshold heuristics); **(iii)** *matched sampling budgets* across stochastic solvers (preventing one-sided advantage); and **(iv)** a *penalty-weight sweep* (revealing the exploitation regime that a single λ would hide). We add a **classical A* baseline** with occlusion-aware cell costs to ground the HUBO formulation against a conventional planner.

Since Qiskit's state-vector simulator scales as 2^n and cannot run at n=960, we use two tiers: **Tier 1** (full 8×8, 960 vars: A*, SA, neal) and **Tier 2** (tiny 2×2, 8 vars: all solvers including QAOA). We use 50 matched samples per stochastic solver, SA at 500 sweeps, and hardened weights λ=1000. A* uses cell cost $1.0 + \lambda_{\text{occ}} \cdot |\text{LOS} \cap O| + \lambda_{\text{prox}} \cdot \mathbf{1}[v_b \in B]$ with Manhattan heuristic.

IV. Results

A. Tier 1 — Three Solvers, One Instance

Table I reports the matched comparison. A* and SA achieve comparable soft cost (317 vs. 303), but A* completes in 10 ms while SA needs 47.8 s — a ∼4,700× speedup with guaranteed feasibility. Neal is fast per sample (0.26 s) but only 22 % of its samples are truly feasible, and its best feasible soft cost (496) is 56 % worse than A*'s. Notably, SA's *mean* soft cost (∼495) matches neal's *best* feasible soft cost (496): neal needs many reads to occasionally equal SA's typical quality. Fig. 2 shows the A* trajectory.

B. The Constraint-Exploitation Curve

Sweeping λ (Fig. 3) reveals the failure mode the methodology targets. At $\lambda \leq 300$ neal achieves 0 % feasibility — every low-energy sample violates constraints. Feasibility rises

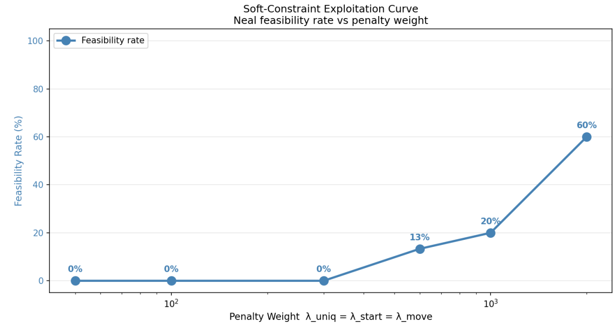

Fig. 3: Constraint-exploitation curve: neal feasibility vs. penalty weight λ. General-purpose annealers need weights an order of magnitude above the objective scale to respect one-hot HUBO structure.

to 13 % at λ=600, 20 % at λ=1000, and 60 % at λ=2000. A single-λ report would mis-state neal's capability.

C. Tier 2 — Three-Way Comparison with QAOA

On the 8-variable instance, A*, SA, neal, and Qiskit QAOA all reach the same optimum $E^*{=}-118.17$ (QuadraticProgram → QAOA, SPSA, reps=1). Runtime is 55 s for QAOA vs. 0.02 s (neal), 0.14 s (SA) — a 2,750× overhead that grows exponentially with qubit count.

V. Discussion and Conclusion

The A* baseline answers the obvious reviewer question: the HUBO captures sensible planning structure — A* finds a feasible trajectory in milliseconds with soft cost within 5 % of the best stochastic solver. The HUBO framing's value is not raw performance on this static-target instance; it is compatibility with quantum-style solvers and extensibility to a *moving* target, where Eq. (2) becomes a joint product of UAV and target occupancy — intrinsically cubic and outside A*'s shortest-path framing. Feasibility-aware benchmarking proves necessary: 78 % of neal's hardened-weight samples still violate constraints. Future work: moving target, real D-Wave hardware, and comparison with the multi-convex MPC of [1].

References

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